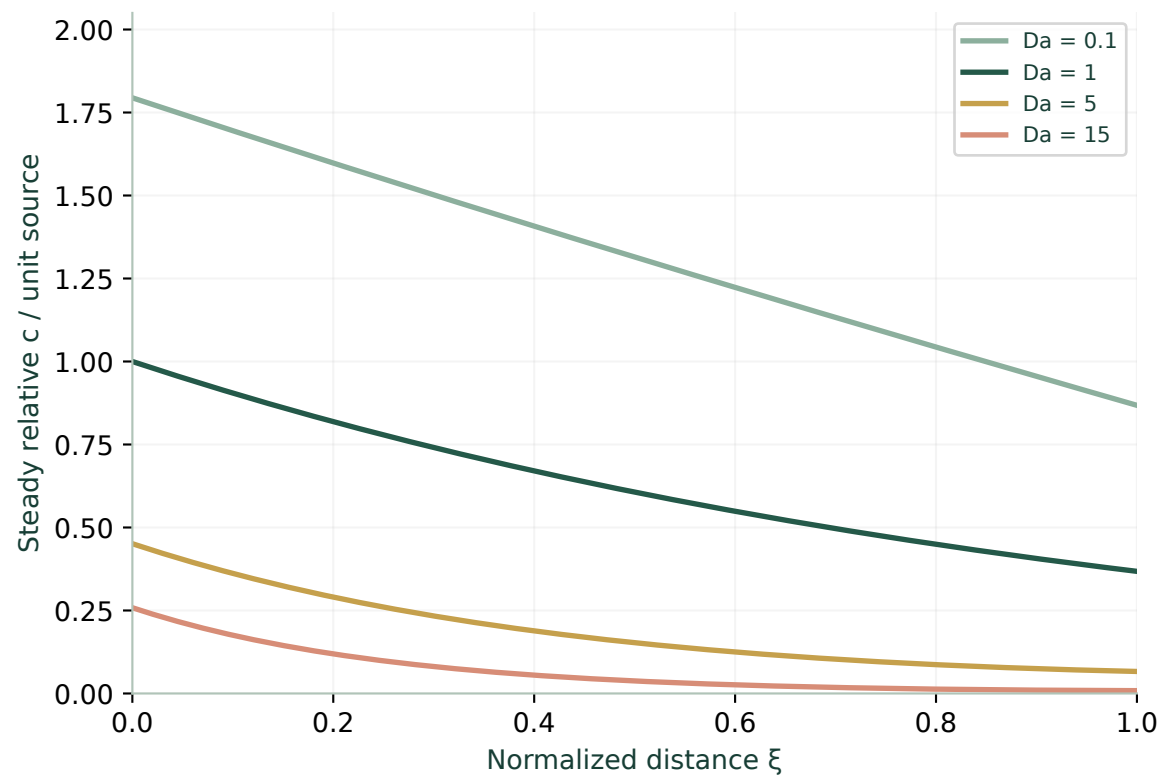
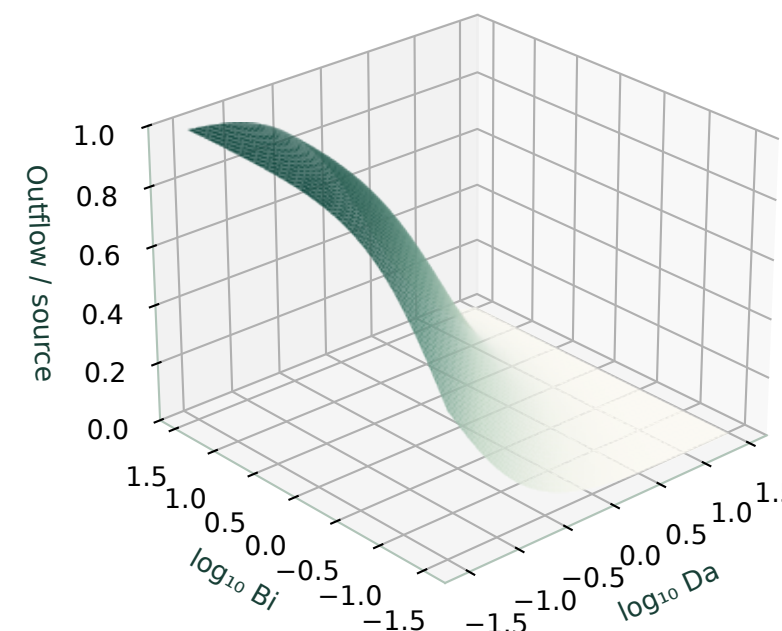


A closed-form benchmark across 10,201 transport conditions

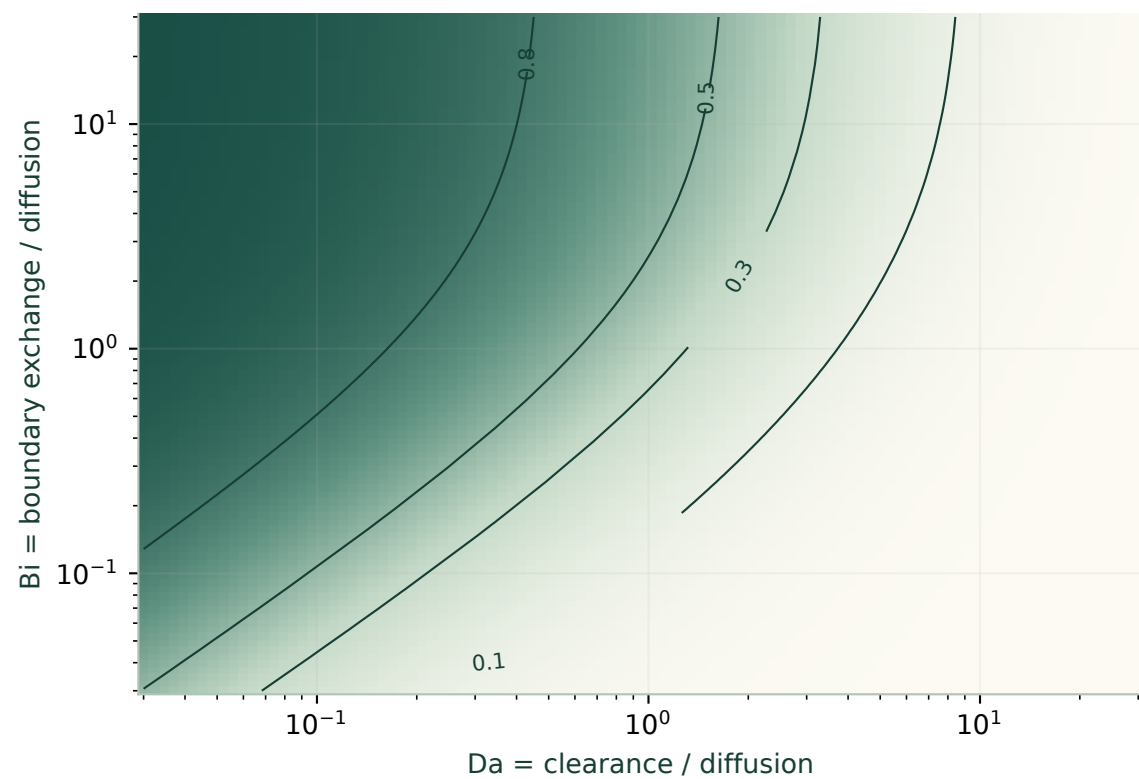
A Clearance shapes the steady profile; $Bi = 1$



B Analytic transport fraction, not efficacy



C Dimensionless groups identify transport regimes



D Time-scale ratio changes the transient, not steady state

